

# Discovery Executive Summary

Project: test-discovery · Generated: 14/07/2026, 18:44:17

EXECUTIVE SUMMARY

This report consolidates the overall ratings, key findings, and recommended actions from the 1 discovery analysis run across this codebase (frontend and backend). Each section below reproduces that analysis's executive view; full evidence and diagrams live in the individual reports.

## Portfolio Overview

| # | Analysis                       | Overall Rating | Hotspot Score |
|---|--------------------------------|----------------|---------------|
| 1 | Architecture & Design Analysis | HIGH RISK      | —             |

## 1. Architecture & Design Analysis

OVERALL CODEBASE RATING — ARCHITECTURE & DESIGN

High Risk

Driven by High-Risk Missing Service Layer (H2), Missing Repository Pattern (H3), Domain Boundary Violations (H8), Shared Database Coupling (H9), and Parallel API Runtimes (H10).

EXECUTIVE SUMMARY

The Klearcom (FSDKC) codebase spans three runtimes — a Laravel 12 API (6 controllers, 4 Eloquent models, 2 services, 0 repositories), a React/Vite/TS SPA (17 source files), and a parallel Node `dev-api` (8 JS files) that mirrors production endpoints. Frontend architecture is comparatively healthy (avg 64 LOC per component, centralized API client, max prop depth 1), but backend layering is the dominant risk: 25 controller-to-model access points bypass a thin service tier, 67 ORM access points occur outside any repository abstraction, and six cross-domain violations couple Discovery and Connect modules through shared models and duplicated tree/reachability logic. All five MariaDB tables are shared across domains with no schema ownership (100% coupling), and two parallel API runtimes (Laravel + dev-api) amplify drift risk. Overall verdict: **High Risk**, driven by H2, H3, H8, H9, and H10.

### 1.1 Benchmark Ratings Summary

| #   | Hotspot                                  | Primary KPI                                   | GOOD | MODERATE | HIGH RISK | Measured                  | Rating    |
|-----|------------------------------------------|-----------------------------------------------|------|----------|-----------|---------------------------|-----------|
| H1  | Fat Controllers                          | Avg LOC per controller                        | <150 | 150–300  | >300      | 62 LOC avg                | GOOD      |
| H2  | Missing Service Layer                    | Controllers accessing repos/models            | <10  | 10–20    | >20       | 25 access points          | HIGH RISK |
| H3  | Missing Repository Pattern               | Direct DB access points                       | <10  | 10–20    | >20       | 67 ORM access points      | HIGH RISK |
| H4  | Circular Dependencies                    | Dependency cycles                             | 0    | 1–3      | >3        | 0                         | GOOD      |
| H5  | Shared Utility Abuse                     | Utility files w/ business logic               | 0    | 1–5      | >5        | 1<br>( LegacyDataMapper ) | MODERATE  |
| H6  | Direct SQL in Controllers                | ORM compliance %                              | >90% | 60–90%   | <60%      | 100% ORM                  | GOOD      |
| H7  | God Classes                              | Classes >1000 LOC                             | 0    | 1–3      | >3        | 0                         | GOOD      |
| H8  | Domain Boundary Violations               | Cross-domain access points                    | 0    | 1–5      | >5        | 6                         | HIGH RISK |
| H9  | Shared Database Coupling                 | Tables shared across domains                  | <10% | 10–30%   | >30%      | 100%                      | HIGH RISK |
| F1  | Business Logic in Components             | Avg LOC per component                         | <150 | 150–300  | >300      | 64 LOC avg                | GOOD      |
| F2  | Missing Frontend Service/Data Layer      | Components w/ inline API calls                | <10  | 10–20    | >20       | 7 files                   | GOOD      |
| F3  | God / Oversized Components               | Components >400 LOC                           | 0    | 1–3      | >3        | 0                         | GOOD      |
| F4  | Prop Drilling / Global State Abuse       | Max prop-drilling depth                       | ≤2   | 3–4      | >4        | 1 level                   | GOOD      |
| F5  | Legacy / Inconsistent Component Patterns | Legacy-pattern components                     | 0    | 1–10     | >10       | 1 class component         | MODERATE  |
| H10 | Parallel API Runtimes (additional)       | Duplicate API implementations across runtimes | 0    | 1        | ≥2        | 2 (Laravel + dev-api)     | HIGH RISK |

## 1.4 Actions Required

| Hotspot                         | Action                                                                                                                                                                                                             | Rating    | Priority |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|
| H2 — Missing Service Layer      | Introduce <code>ConnectMonitorService</code> , <code>DiscoveryJobService</code> , and <code>DashboardKpiService</code> ; move all 25 controller model-access points into application services with constructor DI. | HIGH RISK | CRITICAL |
| H3 — Missing Repository Pattern | Create repository interfaces for all 4 Eloquent models; route 67 ORM access points through injected repository implementations.                                                                                    | HIGH RISK | CRITICAL |
| H5 — Shared Utility Abuse       | Replace <code>LegacyDataMapper</code> <code>extract()</code> pattern with typed DTO mappers; remove <code>extract(\$filters)</code> from <code>LegacyReportController</code> .                                     | MODERATE  | MEDIUM   |
| H8 — Domain Boundary Violations | Split cross-domain access in <code>DashboardController</code> and <code>LegacyReportController</code> ; enforce Discovery/Connect bounded contexts with published read APIs.                                       | HIGH RISK | CRITICAL |
| H9 — Shared Database Coupling   | Assign table ownership per domain in <code>docker/mariadb/init.sql</code> ; introduce per-domain migration files and integration views for cross-domain KPIs.                                                      | HIGH RISK | CRITICAL |
| H10 — Parallel API Runtimes     | Deprecate <code>dev-api</code> duplicate runtime; consolidate to Laravel API with Docker Compose for local dev, or generate dev-api from OpenAPI spec.                                                             | HIGH RISK | CRITICAL |
| F5 — Legacy Component Patterns  | Rewrite <code>LegacyMonitorPoller.jsx</code> as function component with <code>useEffect</code> cleanup; add Error Boundary around <code>LegacyDashboardWidget</code> .                                             | MODERATE  | MEDIUM   |

## 1.5 Expected Outcomes

- **Testable business logic:** Application services and repositories enable unit tests without HTTP or database bootstrapping, cutting test setup time for Connect reachability and Discovery tree workflows.
- **Independent domain evolution:** Bounded contexts with owned schemas let Discovery and Connect teams ship schema changes without cross-module regressions in dashboard KPIs.
- **Single source of truth:** Eliminating the parallel `dev-api` runtime removes behavioral drift between local development and production Laravel deployments.
- **Reduced change amplification:** Centralizing reachability calculation and IVR tree building in one service eliminates the current four-way duplication across controllers, services, and Node handlers.
- **Frontend stability:** Migrating the legacy class component and adding Error Boundaries prevents interval leaks and uncaught render errors during SPA navigation.

### Deliverables saved:

- Full report: [agent-runs/20260714T183917\\_7ofc0a/docs/discovery/01-architecture-design.md](#) (35 KB — includes §1.2 evidence and §1.3 Mermaid diagrams)
- Pipeline summary: [agent-runs/20260714T183917\\_7ofc0a/01-architecture-design-summary.md](#)

Analysis covered **backend** (26 PHP files, 6 Laravel controllers + 12 Express handlers), **frontend** (17 TS/TSX/JSX files), and **dev-api** (8 JS files) from [shende-shweta/FSDKC@main](#) via GitHub REST API.